

Evaluating EmotionFlow on the IEMOCAP Dataset

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Abstract—Emotion recognition in conversation (ERC) is a growing research area critical for human-computer interaction. This study evaluates the EmotionFlow model’s effectiveness on the widely used IEMOCAP dataset. We explore the model’s ability to capture short-range and long-range dependencies, investigate the impact of its Question-Answering (QA) input formatting, and assess the contribution of the Conditional Random Field (CRF) layer. We establish a baseline model and systematically analyze performance improvements provided by each component. Our findings suggest EmotionFlow significantly outperforms simpler baselines, highlighting the effectiveness of modeling conversational context and speaker dynamics.

I. INTRODUCTION

Emotion recognition in conversation (ERC) is a crucial component of building emotionally intelligent AI systems, with applications ranging from human-robot interaction to mental health monitoring. Despite recent progress in deep learning, effectively modeling the emotion shift in multi-turn dialogues remains a challenging task due to the complex dependencies between utterances and speakers. In this work, we conduct a reproduction and extension study of the EmotionFlow model [1], which was originally developed and evaluated on the MELD dataset. Our goal is to assess the generalizability and performance of this model on the IEMOCAP dataset, which offers complementary properties to MELD—including more natural dyadic interactions, a wider range of annotated emotions, and multimodal recordings. In contrast to MELD, which is based on pre-scripted and comedic television dialogue, IEMOCAP features emotionally rich, semi-scripted and improvised exchanges between two speakers in controlled settings. This setup provides more precise utterance segmentation and annotation, making it particularly suitable for evaluating models that depend on contextual and emotional dynamics. We also perform an ablation analysis to investigate the individual contributions of the model’s key components: the question-answering (QA) style input formatting and the Conditional Random Field (CRF) layer. Our study offers new insights into how EmotionFlow performs on IEMOCAP and clarifies the role of its architectural choices.

II. BACKGROUND

A. Related Work

Emotion recognition in conversation (ERC) has evolved significantly over the past decade. Early approaches relied heavily on traditional machine learning models and handcrafted features, which were later replaced by deep learning architectures—most notably, Recurrent Neural Networks (RNNs)

and their gated variants (LSTMs and GRUs). These models enabled the modeling of sequential context and were foundational to early ERC systems. DialogueRNN [2] extended RNN-based architectures by introducing explicit modeling of speaker states and global conversational context. It tracked emotional states at the speaker level using separate recurrent units and demonstrated strong performance on several benchmark datasets.

With the emergence of transformer architectures, ERC models shifted toward attention-based approaches capable of capturing long-range dependencies more efficiently than RNNs. DialogXL [3], built on top of XLNet, introduced utterance-level recurrence and dialog-aware self-attention mechanisms, improving the modeling of multi-party conversations. EmotionFlow builds on this trend by leveraging transformer-based embeddings from RoBERTa [4] and adding a Conditional Random Field (CRF) [5] layer for structured prediction. Unlike other transformer-based models that rely purely on attention mechanisms, EmotionFlow explicitly models emotion transitions across utterances using the CRF.

Recent research has explored incorporating external knowledge—such as commonsense reasoning or affective knowledge graphs—into emotion recognition models to enhance their understanding of implicit emotional cues. These methods aim to supplement raw dialogue data with structured knowledge about typical emotional responses to events, relationships between concepts, or socio-emotional norms. For example, some models integrate commonsense sources like ConceptNet [6] or COMET [7] to improve contextual inference, especially in ambiguous or sarcastic utterances where surface-level semantics may be insufficient.

While these approaches have shown promise in capturing implied emotions and enhancing generalization, they often require careful knowledge selection and introduce additional complexity during training and inference. In contrast, EmotionFlow does not currently incorporate external knowledge bases, focusing instead on learning emotional transitions directly from dialogue structure and speaker context.

B. EmotionFlow

EmotionFlow is a transformer-based model designed specifically to capture the dynamic nature of emotions across dialogues. While many ERC models leverage pre-trained language models such as BERT or RoBERTa to embed individual utterances, EmotionFlow introduces a unique structure by combining RoBERTa-based embeddings with two critical

components: a question-answering (QA) input format and a Conditional Random Field (CRF) output layer.

Unlike standard transformer models that treat each utterance as an isolated input or rely solely on self-attention to capture contextual relationships, EmotionFlow reformulates the emotion classification task as a QA problem. Each input sequence is appended with a query asking for the emotion of the speaker, following relevant conversational history. This structure improves the model’s ability to focus on contextually relevant information and interpret the emotional trajectory of the conversation. The QA formulation reframes the task as contextual reasoning, encouraging deeper understanding beyond flat classification.

The final CRF layer provides structured prediction by modeling dependencies between consecutive emotion labels. While typical softmax classifiers assume independence between predictions, the CRF layer captures label transitions, enabling the model to enforce smoother and more coherent emotional state progressions. This is especially important in dialogue, where emotion shifts are often gradual or dependent on speaker interactions.

In summary, EmotionFlow distinguishes itself from other transformer-based ERC models by (1) explicitly guiding attention through a QA-style format and (2) modeling inter-utterance label dependencies via a CRF layer. Together, these mechanisms allow EmotionFlow to outperform both conventional RoBERTa-based classifiers and context-aware transformers like DialogXL, which lack structured output modeling.

III. METHODS

A. Datasets

IEMOCAP Dataset: The IEMOCAP (Interactive Emotional Dyadic Motion Capture) dataset is widely used for emotion recognition research. Five female and five male actors were selected, asked to rehearse under the supervision of a director who made sure the intended emotions were genuinely expressed, avoiding exaggeration or caricature of the emotions. Participants improvised scenarios based on certain prompts designed to elicit specific emotions—frustration, sadness, happiness, anger, or neutrality. In this setting, the subjects were free to use their own words to express themselves. The actors were given significant freedom to express emotions naturally and authentically. [8]

MELD Dataset: The MELD dataset is an extension of the EmotionLines dataset [9] and includes multimodal, multi-party dialogues extracted from the TV series “Friends”. Each utterance is annotated with emotions (anger, disgust, sadness, joy, neutral, surprise, fear) [10]. MELD contains multi-party dialogues, which makes emotion recognition more difficult due to frequent speaker changes and overlapping interactions. Additionally, since the conversations are taken from a scripted TV show, they may not fully reflect the natural flow and emotional nuance of real-life conversations, potentially limiting the realism and generalizability of the dataset.

Figures 1 and 2 illustrate how emotion labels transition between consecutive utterances. These visualizations highlight

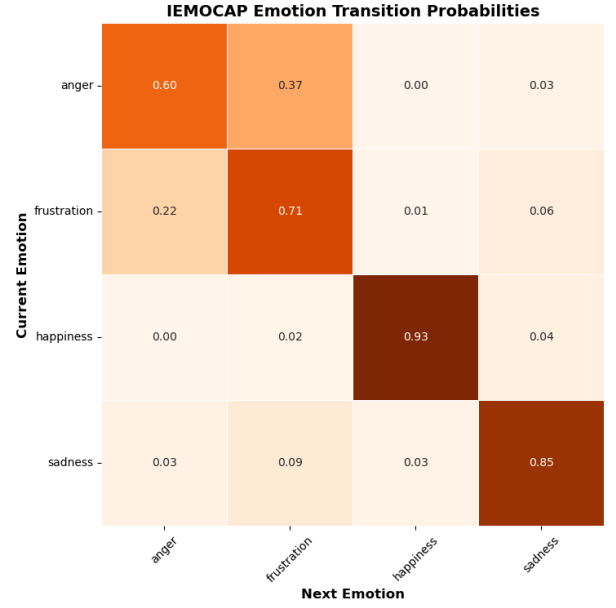


Fig. 1: Transition probability of emotion labels from previous to current utterance in the IEMOCAP dataset.

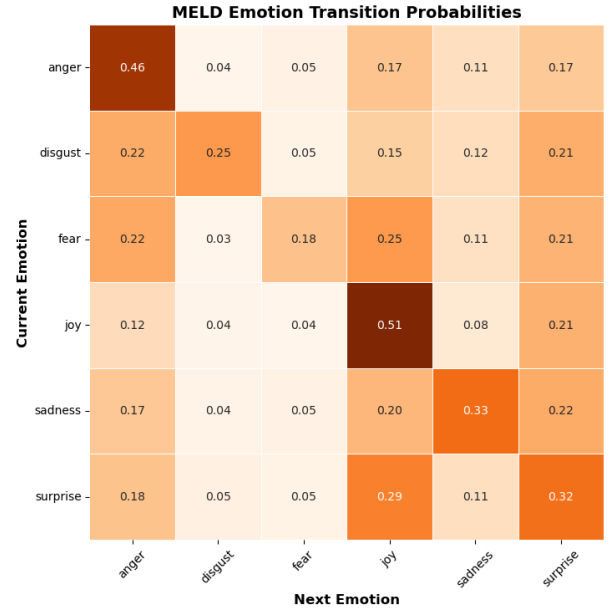


Fig. 2: Transition probability of emotion labels from previous to current utterance in the MELD dataset.

a key characteristic of conversational emotion dynamics: each line in a dialogue influences the emotional tone of the following utterance. This supports the idea that emotions in conversations are not isolated but evolve in response to the preceding context, reinforcing the importance of modeling sequential dependencies in ERC systems.

B. Data Engineering

The raw textual utterances from the IEMOCAP dataset require preprocessing to fit the RoBERTa model input format. We tokenize the data using RoBERTa’s tokenizer, converting

each sequence into token IDs and attention masks. To preserve dialogue structure, we construct the input sequences by concatenating the three most recent utterances prior to the target utterance, followed by the target utterance itself. Each utterance is prefixed with the speaker ID, for example: “Speaker 1 says: How are you?” A query is then appended in the format: “Speaker 1 feels $\langle \text{mask} \rangle$.” This format explicitly incorporates conversational context and speaker identity, guiding the model to infer the emotional state of the current speaker. This structured preprocessing enables the EmotionFlow model to more effectively model contextual dependencies within multi-turn dialogues.

Compared to the MELD dataset, which contains 13,708 utterances across 1,432 dialogues (approximately 10 utterances per dialogue), shown in Table I, the IEMOCAP dataset consists of 10,087 utterances distributed across just 151 dialogues—an average of approximately 67 utterances per dialogue, shown in Table II. Training on such lengthy dialogues significantly increases computational resource demands and model memory usage during fine-tuning.

TABLE I: MELD Dataset Statistics

	Train	Dev	Test	Total
Conversations	1038	114	280	1432
Utterances	9989	1109	2610	13708
Speakers	260	47	100	274

TABLE II: IEMOCAP Dataset Statistics Before Engineering

	Train	Dev	Test	Total
Conversations	108	12	31	151
Utterances	7160	783	2144	10087
Speakers	216	24	62	302

TABLE III: IEMOCAP Dataset Statistics After Engineering

	Train	Dev	Test	Total
Conversations	820	92	229	1141
Utterances	7244	803	2040	10087
Speakers	302	146	232	680

To address this, a preprocessing step was included to split long IEMOCAP dialogues into smaller units averaging around 9 utterances each, resulting in a new total of 1,141 dialogues, shown in Table III. This segmentation strategy improves training efficiency while preserving the original benefits of the IEMOCAP dataset, including high-quality emotion labels and dyadic interaction structure. Previously, each full-length dialogue involved two speakers, with speaker features isolated to their specific dialogue. By splitting these dialogues, we can now model speaker context, as speakers may now appear across more numerous and consistent dialogue segments, enabling more robust tracking of speaker-specific emotional dynamics.

C. Implementation

To evaluate the impact of different architectural components in the EmotionFlow model, we implemented a series of model variants, each building on RoBERTa as the shared embedding backbone, with specific modifications to isolate

the contributions of the QA input formatting and the CRF-based sequence modeling. By systematically comparing these variants, we aim to assess the role of contextual representation and label transition modeling in improving performance on conversational emotion recognition tasks. Four model variants were trained for evaluation:

- 1) **Baseline Model:** RoBERTa embeddings followed by a linear classification head that outputs raw logits. Input sequences are constructed by concatenating the three preceding utterances with the current one to provide short-term context. Final predictions are obtained by applying argmax over the logits.
- 2) **EmotionFlow w/o QA:** EmotionFlow architecture without the question-answering (QA) input format, removing the auxiliary emotional query for context grounding while retaining the CRF layer for sequential modeling.
- 3) **EmotionFlow w/o CRF:** This variant uses the same architecture as the baseline model—RoBERTa embeddings followed by a linear classification head—but incorporates QA-style input formatting. It does not use a CRF layer.
- 4) **Full EmotionFlow:** The original EmotionFlow model using QA-based input sequences and a CRF layer to model dependencies between consecutive emotion labels.

The baseline and non-CRF variants are trained using cross-entropy loss for multi-class classification. The full EmotionFlow model, which includes the CRF layer, is trained using the negative log-likelihood loss from the CRF.

D. Training and Hyperparameter Tuning

We conducted a series of training experiments to evaluate the impact of key hyperparameters on model performance. All models were trained using the Adam optimizer, with learning rates tuned between $1\text{e-}4$ and $2\text{e-}4$ for the main task and $1\text{e-}5$ for pre-trained layers. The initial experiment (Experiment 1) used a small batch size of 4 and short maximum sequence length (32 tokens) for debugging and assessing training stability. Subsequent experiments adjusted parameters to improve learning dynamics and model capacity.

In Experiment 2, we increased the sequence length to 128 while reducing the batch size to 1 (accumulated over 8 steps) to manage memory usage, and extended training to 5 epochs. This configuration served as a baseline for comparing model variants. Experiment 3 introduced a warmup phase (2 epochs), a higher learning rate ($2\text{e-}4$), and increased dropout to 0.4, yielding improved performance across IEMOCAP and MELD, especially for models with QA formatting and CRF. Experiment 4 maintained the improved learning rate and warmup settings but extended training to 8 epochs and reverted dropout to 0.3. This led to further improvements on MELD, although it caused overfitting on IEMOCAP. Results below show the best F1 score on the development set from experiment 3 for IEMOCAP (Table IV) and from experiment 4 for MELD (Table V).

The QA-formatting and CRF-based EmotionFlow variants consistently outperformed simpler models. Hyperparameter

tuning was conducted manually by monitoring the validation set weighted average F1 score trend.

E. Evaluation and Metrics

To evaluate model performance, we use the weighted average F1 score, which balances precision and recall across emotion categories while accounting for class imbalance—a known issue in emotion-labeled datasets like IEMOCAP and MELD. Weighted F1 is preferred in this task because it captures overall model effectiveness without being disproportionately influenced by dominant emotion classes (e.g., neutral). This metric is widely used in ERC literature, including prior evaluations of EmotionFlow and competing models such as DialogueRNN and DialogXL.

We define model success as achieving a higher weighted F1 score than the baseline model, and demonstrating consistent improvement through the inclusion of either the QA formatting or CRF layer. This ties directly back to our research objective: to evaluate whether modeling emotional transitions and conversational context yields measurable improvements in emotion recognition.

IV. RESULTS AND DISCUSSION

Table IV summarizes the F1 scores for the IEMOCAP dataset, while Table V presents the F1 scores for the MELD dataset. These tables illustrate the comparative performance of the EmotionFlow model variants, including baseline and ablated models. Across both datasets, the Full EmotionFlow model consistently outperforms simpler architectures, highlighting the benefit of combining the QA-style input with CRF-based sequence modeling.

Model	Best Dev F1 Score
Baseline (RoBERTa only)	65.98
EmotionFlow w/o QA	66.13
EmotionFlow w/o CRF	68.33
Full EmotionFlow	69.42

TABLE IV: F1 Scores for IEMOCAP Dataset

Model	Best Dev F1 Score
Baseline (RoBERTa only)	61.06
EmotionFlow w/o QA	62.51
EmotionFlow w/o CRF	62.10
Full EmotionFlow	62.90

TABLE V: F1 Scores for MELD Dataset

Our initial experiments with the baseline model showed that RoBERTa alone was insufficient to capture emotional dynamics effectively. Incorporating the QA-based input format improved the model’s ability to focus on relevant context. Similarly, the CRF layer provided smoother transitions between emotional states across utterances, leading to more stable and coherent predictions.

Figure 3 shows the development F1 scores over epochs for the IEMOCAP dataset, while Figure 4 shows the development F1 scores over epochs for the MELD dataset. While both curves exhibit improvement, IEMOCAP’s F1 curve shows greater volatility, particularly in earlier epochs. This aligns

with our observation that longer, naturalistic dialogues in IEMOCAP are more emotionally complex and require additional epochs for the model to generalize well. Researchers working on MELD and IEMOCAP have similarly reported that MELD’s structured scenes lead to faster initial convergence, while IEMOCAP requires finer modeling strategies to stabilize and improve performance.

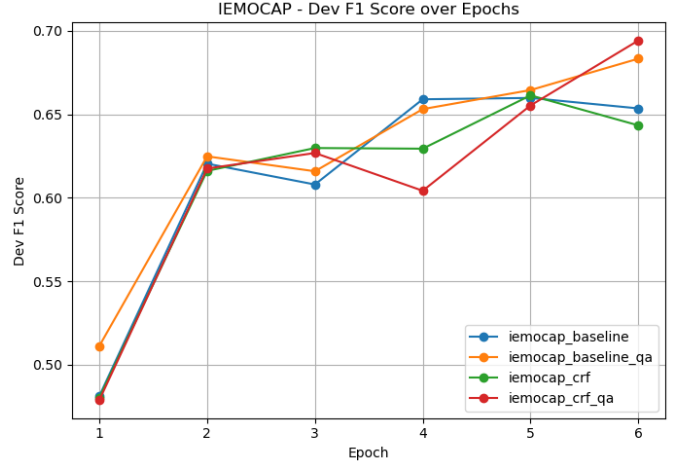


Fig. 3: Dev F1 Scores Over Epochs (IEMOCAP)

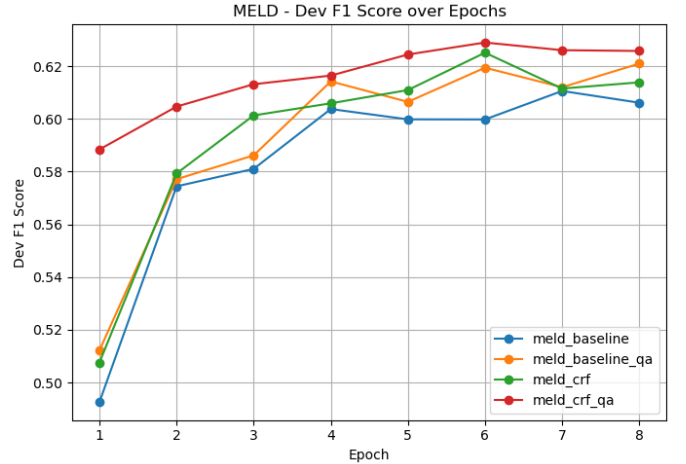


Fig. 4: Dev F1 Scores Over Epochs (MELD)

Overall, our results confirm that EmotionFlow’s architecture—with its combination of QA-style contextual queries and CRF-based label smoothing—provides consistent performance gains on both structured and naturalistic datasets. These findings mirror results reported in the original EmotionFlow paper on MELD, while extending their validation to a more challenging and dynamic dataset in IEMOCAP.

However, we were not able to fully reproduce the MELD performance reported in the original paper. While the original EmotionFlow model achieved a dev F1 score of 65.05 on MELD, our best result was 62.90. This performance gap may be due to limited computational resources, which restricted our ability to use larger batch sizes, longer token lengths, and longer training schedules. Despite this, our results still validate

the architectural benefits of EmotionFlow across different conversational contexts.

V. CONCLUSION

In conclusion, our reproduction and extension of the EmotionFlow model on the IEMOCAP dataset not only reinforce the model’s effectiveness on structured, multi-party conversations like MELD, but also demonstrate its ability to generalize to emotionally rich, dyadic dialogues. Our experiments showed that the full EmotionFlow model—combining QA-style input formatting with CRF-based sequential modeling—consistently outperformed ablated variants and a strong contextual baseline on both MELD and IEMOCAP. While we were unable to fully replicate the original MELD performance reported in the EmotionFlow paper, our results still demonstrate clear gains from incorporating structured prediction and contextualized queries. The ablation analysis confirmed that both the QA and CRF components contribute meaningfully to performance, particularly on more emotionally complex dialogues like those in IEMOCAP.

Additionally, by engineering IEMOCAP to support shorter, context-preserving dialogue segments, we maintained the dataset’s expressive richness while enabling finer-grained speaker modeling and more stable training dynamics. Overall, our findings highlight EmotionFlow’s adaptability across different conversational structures and suggest that structured input and output strategies are crucial for advancing emotion recognition in dialogue systems.

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